

CALCULATING YOUR EMISSIONS PROFILE

The amount of carbon emissions that each of us is responsible for, and the sources of those emissions, may surprise you.

BY SHUI BIN

Most people—even those who are motivated to become more environmentally responsible—are not very familiar with the energy- and carbon-related impacts of their day-to-day activities. Some people think that turning off lights, recycling cans, and becoming vegetarians are the most significant choices they can make. Although these behaviors often reduce energy use and CO₂ emissions, their overall effect is minimal. Meanwhile, unwitting persistence in other activities, such as frequent air travel and using electricity as a source of heat, can lead to significant energy use and CO₂ emissions. Well-designed and credible consumer information campaigns are needed to help consumers make better-informed and more-effective choices.

As a first step toward making effective choices, consumers need to be informed about the energy use and the related environmental impacts of their activities; in short, each consumer needs to understand his or her own profile of CO₂ emissions.

Several tools are available to help people acquire that knowledge. Home Energy Saver (<http://hes.lbl.gov>), developed by Lawrence Berkeley National Laboratory, is an online tool that demonstrates energy use and expense by end uses in U.S. homes. An online personal CO₂ calculator (www3.iclei.org/co2/co2calc.htm), developed by the International Council for Local Environmental Initiatives, calculates the CO₂ emissions from home



MARY JAMES

Steve Greenberg, *Home Energy's* technical editor, would rather let his clothes flap in the breeze than use a clothes dryer.

energy use and personal travel. The Climate Change Calculator (www.climatecalc.net) is an online interactive tool designed to raise Canadians' awareness of the greenhouse gases that they produce through their lifestyle choices.

All of these tools estimate the energy use or CO₂ emissions that are directly associated with a consumer's actions. The Individual CO₂ Emissions Estimator, or ICEE, is a stand-alone software program that I designed that goes one step further. This program reflects CO₂ emissions that are both directly and indirectly influenced by consumer activities. The direct influences in the current

version of ICEE include home energy use and personal travel; the indirect influences include food consumption, car purchases, and recreational activities.

In designing ICEE, I tried to include enough behavioral variables so that the CO₂ emissions resulting from consumers' purchase choices and use behaviors would be fairly modeled. I assumed that the more personalized a tool can be, the more likely it is that users will accept it as a reflection of their reality; and that the larger the number of behavior variables, the better the chances of developing a good experimental tool for the study of individual choices.

Examples of Emissions Profiles

Few people have an exact sense of how much CO₂ they emit in a year. I conducted a small study to find out how people respond who are given this information, and if the information leads them to make more environmentally responsible choices. Three people from Pittsburgh participated in my



Liz Varnhagen cultivates CO₂-consuming plants in her and Steve Greenberg's garden.

MARY JAMES

study. The ICEE was used to calculate each of their CO₂ emissions profiles. The first participant was a white man who works at UniMart as a clerk. He is in his mid-20s and lives alone. His annual income is below \$15,000. The second participant was a 20-something African American woman with a graduate degree. She lives with her mother. Both mother and daughter are employed in white-collar jobs. Their household income is about \$65,000 per year. The third participant was a 40-something white

WHAT YOU CAN DO TO REDUCE CO₂ EMISSIONS

Living in a small house or apartment, rather than in a big house and driving an energy-efficient car rather than an SUV are obvious ways that you can reduce CO₂ emissions. And practice the three Rs: Reduce how much you use, Reuse what you can, and then Recycle the rest and dispose of what's left in the most environmentally friendly way.

Here are some more ways to save the earth:

- Do everything suggested in the pages of *Home Energy* that is appropriate to your living situation, such as repairing leaky ducts, improving windows, and having enough insulation for your climate. (For a helpful tool in studying your home's energy use, see <http://hes.lbl.gov>.)
- Purchase and use Energy Star appliances. (For a listing of Energy Star appliances, see www.energystar.gov.) Use energy-efficient lighting and fixtures.
- Adjust your clothing level to expand your comfort envelope, and take advantage of it by turning up your air conditioning thermostat in summer (78°F) and turning down your heating thermostat in winter (68°F). Set back the heat even more and set up the cooling even higher at night, and give your heating and cooling systems a rest when you are out of the house.
- Turn off lights when they are not in use. Many appliances, such as

televisions, microwave ovens, and cell phone chargers, use energy even when they are switched off. Unplug these appliances when they are not in use. This is easy to do if you plug several appliances into one power strip.

- If you have access to both electricity and natural gas, use gas to power your water heating, space heating, and cooking (unless water and space heating are provided by heat pumps).

- Use your clothes dryer sparingly. Invest in a clothesline and use it to dry your clothes and sheets—it'll save energy while making your clothes and sheets smell fresh. Use laundry detergent that cleans well in cold water.

- Use green energy, such as solar and wind, to power your home and generate hot water, or purchase renewable energy credits to offset your home's CO₂ emissions (see "Clean Up Your Ecological Footprint," p. 12).

- Think about cutting back on the meat that you and your family consume. A lot more energy goes into getting that pound of hamburger to your dinner table than to producing a loaf of whole wheat bread.

- Water takes energy to treat and transport both upstream and downstream of the point of use. Save water and energy whenever possible with low-flow toilets (see "Which Toilets Deliver?" p.19), low-flow shower fixtures, and by planting native and drought-resistant plants and

grasses in your yard instead of plants and grasses that require a lot of watering (see "Tucson Blooms with Less Water," *HE* May/June '04, p. 15).

- Purchase and use vehicles that can run on biodiesel, electricity, ethanol, hydrogen, natural gas, propane, or a mixture of gasoline or diesel and a renewable fuel. If you choose to use a gasoline-powered vehicle, make sure that it has high fuel efficiency. Consider purchasing a hybrid.

- Take good care of your car so that it operates efficiently. Keep your tires inflated to the correct pressure. Cars are designed to be most efficient at just below the typical highway speed limit. Avoid speeding tickets and excess emissions by driving at or below the speed limit. Also, avoid rapid and unnecessary starts and stops. That means don't tailgate; it's safer and is much less likely to elicit road rage from the people you are following.

- Drive less, bike and walk more, carpool, and use public transportation whenever possible. If your job allows it, try telecommuting one or more days a week.

- A big emissions producer for many people is air travel. You can mitigate the effects of air travel through renewable energy credits.

—Jim Gunshinan

Jim Gunshinan is Home Energy's managing editor.



MARY JAMES

Steve and Liz forego driving gas-guzzling vehicles in favor of a two-wheeled means of transportation.

woman. She and her husband have three children. Their household annual income is below \$50,000.

All three participants were asked to give household demographic information, such as the number of household members, and, using ICEE, to fill in a questionnaire on household consumption related to home energy, personal travel, food, and entertainment. After a participant clicked a calculation button, ICEE presented his or her annual CO₂ emissions distribution by consumption activities. ICEE also provides the average CO₂ emissions for a single person in the United States for comparison purposes. Then participants were asked what choices they would like to make in light of their estimated annual CO₂ emissions. The impacts of their chosen changes were translated into calculated emissions.

Participant 1 lives in a small row house without cooling. He uses natural gas for space heating, water heating, and cooking. His dryer is powered by electricity. He usually takes a bus to work, logging about 5,200 miles per year. He has no car. He spends about \$20 a week on food. He spends \$500 a year on magazines and books. His annual CO₂ emissions total 2.8 tons, only 30% of the total for the average U.S. consumer. About half of his energy consumption is attributable to his home energy use.

Participant 2 and her mother live in a large single-detached house. They use

natural gas for space heating. Room temperature is kept at 76°F during heating seasons, and 72°F during cooling seasons. Their water heater, dryer, and cooking appliances are fueled by electricity. They watch 25 hours of TV each week. The household has more than 20 100W incandescent light bulbs, each of which is used less than four hours a day. The household travels 14,000 miles per year by bus and car; 85% of those miles are used for commuting. They travel about 7,200 miles by air, all for leisure purposes. They spend \$100 a week on food and about \$200 a year on books, magazines, and newspapers. The annual CO₂ emissions per household member total 12.4 tons, 38% higher than the U.S. average. Personal travel (6.8 tons) and home energy (5.2 tons) are the main contributors.

Participant 3 lives with her large household in a medium-sized single-detached house, with no cooling in the summer. The household uses gas for space heating, water heating, and cooking. They keep the indoor temperature at 68°F during heating seasons. They have an automatic-defrost side-by-side refrigerator, which they bought after 1995. They watch about 45 hours of TV a week. They have 14 100W incandescent light bulbs, each of which is on for four to eight hours a day. They have two cars, which they drive for a total of about 24,000 miles per year. Half of

those miles are used for commuting. They spend about \$220 a week on food. They spend \$900 a year on pet products and services and \$250 on recreational books, magazines, and newspapers. The annual CO₂ emissions per household member total 7.7 tons, about 85% of the U.S. average.

Personal Choices

When asked what changes he might be willing to make, Participant 1 claimed that he had no reason to make changes because his emissions rate was well below the average. He believes that his current level of emissions is the minimal level for an American.

After learning that her CO₂ emissions were well above the U.S. average, Participant 2 appeared eager to bring her emissions down to the average level. She attempted to reduce CO₂ emissions by taking fewer showers, washing the dishes less often, and watching less TV. She considered reducing weekly expenditures on restaurant food and on recreation. However, all these choices will reduce her overall emissions by only 2.5%. Even after these changes, her emissions are still above the U.S. average. When Participant 2 realized this, she was upset.

Participant 3 said she was planning to increase the frequency with which she washed clothes—from 8 times to 11 times per week. This will increase her annual CO₂ emissions by 0.1 ton, an increase of 1%.

Lessening the Impact

Although all three participants expressed great interest in the results generated by the ICEE, they didn't all express a desire to reduce their carbon emissions. Participant 1 said clearly that he did not intend to try to reduce his emissions, and Participant 3 planned to wash her family's clothes more often. These results support earlier research, which found that feedback was effective in reducing consumption for high consumers, but that it may have the opposite effect for low and medium consumers. Participant 1 is an example of a low con-

sumer and Participant 3 is an example of a medium consumer.

Participant 2 was interested in reducing her level of emissions, but she seemed not to know how to do so. As the pilot study shows, all of her strategies for being environmentally responsible involved lifestyle sacrifices. None of these sacrifices—cutting down on showers, watching less TV, and eating less restaurant food—had a significant impact. She used electricity for water heating and cooking but she was unaware that she could use natural gas instead. She could have reduced her emissions dramatically by switching to a gas-fired water heater and gas cooking appliances, and she wouldn't have had to make any hard sacrifices.

This pilot study was so small that the results and conclusions presented here should be interpreted and generalized cautiously. But these results do suggest that people still lack accurate information on actions that they can take to reduce carbon emissions, such as fuel switching (from electricity to

natural gas) for dual-fuel home services such as water heating, space heating, clothes drying, and cooking. (For more suggestions, see “What You Can Do to Reduce CO₂ Emissions,” p. 16.) Research from the '80s found that people think of energy conservation in terms of visible curtailment actions, such as turning off lights, watching less TV, and using less hot water. Certainly Participant 2 thought this way.

In our recent study to determine how consumer activities contribute to carbon emissions, Professor Hadi Dowlatabadi and I found that services and products related to the home and personal transportation are prime targets for energy conservation and carbon mitigation. In contrast, the reduction potential of food and clothing consumption is relatively limited. Educational campaigns are needed to let people know which reduction strategies are most effective.



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FOR MORE INFORMATION:

The ICEE is currently being upgraded. The new version of this program should be online by August 2005 at www.isse.ucar.edu/icee.

For more on the research cited in this article, see Bittle, R. and R. Valesano (1979-1980). “The Effects of Daily Feedback on Residential Electricity Usage as a Function of Usage Level and Type of Feedback Information.” *Journal of Environmental Systems* 9(3): 275-287.

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